DOC-ANALOG-001

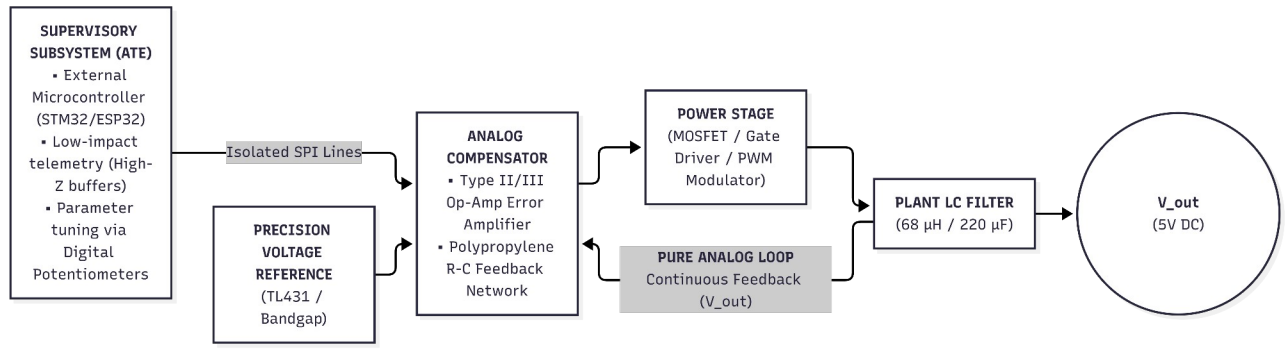
1. Summary and Architectural Philosophy

The fundamental purpose of this project is to revive the purity of continuous analog computing as applied to real-time power control, emulating the philosophy of aerospace guidance and dynamic simulation computers.

Unlike modern digital controllers based on microcontrollers or DSPs, which are inherently limited by:

- *Sampling Latency:* Conversion times of analog-to-digital converters (ADCs).
- *Sampling and Quantization Jitter:* Phase Noise and Finite Resolution in the Error Signal.
- *Computational delay:* Lags caused by interrupt service routines (ISR) and digital PWM refresh rates.

A pure analog control loop processes the error signal using the flow of electrons in the continuous domain. The response speed is limited only by the gain-bandwidth product (GBW) and the slew rate of the selected precision operational amplifiers.



2. Mathematical Modeling and Plant Sizing

The physical design consists of a step-down (Buck) DC-DC converter operating in Continuous Conduction Mode (CCM). Its function is to step down a nominal 24V industrial bus voltage to a stabilized 5V output voltage, supplying up to 2.0A of direct current to the load.

2.1. Nominal Operating Conditions

Parameter	*Symbol*	*Nominal Value*	*Tolerance/Physical Criteria*
Input Voltage	V_{in}	24.0 V	DC bus with typical ripple ±10 %
Output Voltage	V_{out}	5.0 V	Static tolerance < ±0.5 %
Maximum Load Current	I_{out_max}	2.0 A	Rated Design Current
Minimum Load Current	I_{out_min}	291 mA	Critical threshold for preventing DCM
Switching Frequency	f_{sw}	100 kHz	Period T_{sw} = 10 µs
Relative Current Ripple	γ	30 %	$\Delta I_L = 0.30 * I_{out} = 0.60$ A
Output Voltage Ripple	ΔV_{out}	≤ 20 mV_{p-p}	< 0.4 % de V_{out}

2.2. Calculation of Power Inductance

Under steady-state conditions and assuming ideal components, the duty cycle D is defined as:

$$D = \frac{V_{out}}{V_{in}} = \frac{5.0V}{24.0V} \approx 0.20833 \ (20.83\%)$$

During the main switch's conduction interval ($t_{on} = D \cdot T_{sw} = 2.0833\ \mu s$), the voltage applied to the inductor is $V_L = V_{in} - V_{out} = 19.0\ V$. Applying Faraday-Lenz's Law to the volt-second relationship:

$$V_L = L \cdot \frac{\Delta I_L}{t_o n} \Rightarrow L = \frac{(V_{in} - V_{out}) \cdot D}{\Delta I_L \cdot f_{sw}}$$

Substituting the target values:

$$L_{theoretical} = \frac{(24.0 - 5.0) \cdot 0.20833}{0.60 \cdot 100 \times 10^3} = \frac{3.9583}{60000} \approx 65.97 \mu H$$

Seleccionamos el valor comercial estandarizado superior: $L = 68\ \mu H$.

Recalculation of Dynamic Parameters Using L = 68 µH:

1. Real Current Ripple (ΔI_L):

$$\Delta I_L = \frac{(24.0 - 5.0) \cdot 0.20833}{68 \times 10^{-6} \cdot 100 \times 10^3} \approx 0.5824 A_{p-p}$$

2. Peak Current in Inductance (I_{L_peak}):

$$I_{L.peak} = I_{out.max} + \frac{\Delta I_L}{2} = 2.0A + 0.2912A = = 2.2912A$$

3. Magnetic Saturation Margin (I_{sat}):

To prevent a catastrophic drop in inductance caused by saturation of the ferrite core during load transients, the selected physical component must meet the following requirements:

$$I_{sat} \geq 1.3 \times I_{L.peak} = 1.3 \times 2.2912A \approx 2.98A \Rightarrow I_{sat} \geq 3.0A$$

4. Threshold for Transition to Intermittent Driving Mode (DCM):

$$I_{out.crit} = \frac{\Delta I_L}{2} = \frac{0.5824\ A}{2} = 291.2\ mA$$

Design Note: If $I_{out} < 291.2\ mA$, the plant transitions from a second-order system to a first-order system dependent on load resistance R_L, by changing the position of the loop's ends.

2.3. Calculation of the Output Capacitor and ESR Budget

The total voltage ripple at the output node ΔV_{out} is the vector sum/superposition of the pure capacitive component (ΔV_C) and the resistive drop across the Equivalent Series Resistance (ΔV_{ESR}):

$$\Delta V_{out} \approx \Delta V_C + \Delta V_{ESR}$$

To meet the requirement for high analog purity ($\Delta V_{out} \leq 20\ mV_{p\text{-}p}$), we break down the switching noise budget as follows:

- Capacitive Budget (ΔV_C): ≤ 5.0 mV
- ESR by Budget (ΔV_{ESR}): ≤ 15.0 mV

1. Minimum Capacity (C_{out_min}):

$$\Delta V_C = \frac{\Delta I_L}{8 \cdot C_{out} \cdot f_{sw}} \Rightarrow C_{out.min} = \frac{\Delta I_L}{8 \cdot f_{sw} \cdot \Delta V_C}$$

$$C_{out.min} = \frac{0.5824\ A}{8 \cdot (100 \times 10^3\ Hz) \cdot (5 \times 10^{-3}\ V)} = \frac{0.5824}{4000} \approx 145.6\ \mu F$$

We select a market value of C_{out} = 220 µF (Ultra-low ESR aluminum polymer electrolytic capacitor in parallel with a ceramic capacitor 10 µF X7R/C0G to attenuate high-frequency spikes).

2. Maximum Equivalent Series Resistance (ESR_{max}):

$$ESR_{max} = \frac{\Delta V_{ESR}}{\Delta I_L} = \frac{15.0 \times 10^{-3}\ V}{0.5824\ A} \approx 25.75\ m\Omega$$

It is a strict rule that the capacitive output network must ensure an ESR ≤ 25.0 mΩ.

Recalculation of the Expected Total Stress Curvature:

$$\Delta V_C = \frac{0.5824}{8 \cdot 100 \times 10^3 \cdot 220 \times 10^{-6}} = 3.31\ mV$$

$$\Delta V_{ESR} = 0.5824\ A \cdot 0.025\ \Omega = 14.56\ mV$$

$$\Delta V_{out.total} = 3.31\ mV + 14.56\ mV = 17.87\ mV_{p-p}\ (\leq 20\ mV_{p-p})$$

3. Plant Transfer Function

Using the State-Space Averaging (SSA) method developed by R. D. Middlebrook and S. Ćuk, the small-signal transfer function between the duty cycle *d(s)* and the output voltage $v_{out}(s)$ is given by:

$$G_{vd}(s) = \frac{\hat{v}_{out}(s)}{\hat{d}(s)} = V_{in} \cdot \frac{1 + sC_{out}ESR}{1 + s\left(\frac{L}{R_L} + C_{out}(ESR + R_L||ESR)\right) + s^2 LC_{out}\left(1 + \frac{ESR}{R_L}\right)}$$

Since $R_L = V_{out} / I_{out_max}$ = 5 V / 2 A = 2.5 Ω >> ESR = 25 mΩ, the equation simplifies to the canonical second-order form:

$$G_{vd}(s) \approx V_{in} \cdot \frac{1 + \frac{s}{w_z}}{1 + \frac{s}{Qw_0} + \frac{s^2}{w_0^2}}$$

3.1. Location of Poles and Zeros in an Open-Loop System

a) Natural Resonance Frequency of the LC Filter (f_0):

$$w_0 = \frac{1}{\sqrt{L \cdot C_{out}}} = \frac{1}{\sqrt{68 \times 10^{-6} \cdot 220 \times 10^{-6}}} \approx 8164.96 \; rad/s$$

$$f_0 = \frac{w_0}{2\pi} \approx 1300.0 \; Hz \; (1.30 \; kHz)$$

b) Zero Introduced by the Capacitor's ESR (f_z):

$$w_z = \frac{1}{C_{out} \cdot ESR} = \frac{1}{220 \times 10^{-6} \cdot 25 \times 10^{-3}} = 181818.18 \; rad/s$$

$$f_z = \frac{w_z}{2\pi} \approx 28952.4 \; Hz \; (28.95 \; kHz)$$

c) Resonance Quality Factor (Q):

$$Q \approx \frac{R_L}{w_0 L} = \frac{2.5}{8164.96 \cdot 68 \times 10^{-6}} \approx 4.50 \; (+13.07 \; dB \; resonance \; peak)$$

Physical Interpretation of Control: At a frequency of 1.30 kHz, the LC filter introduces a double pole that causes an abrupt drop in gain at a rate of -40 dB/decade and a critical phase drop of -180º. The analog compensator must provide a substantial phase boost near the crossover frequency to prevent unstable oscillation of the converter.

4. Physics of Discrete Components and Non-idealities

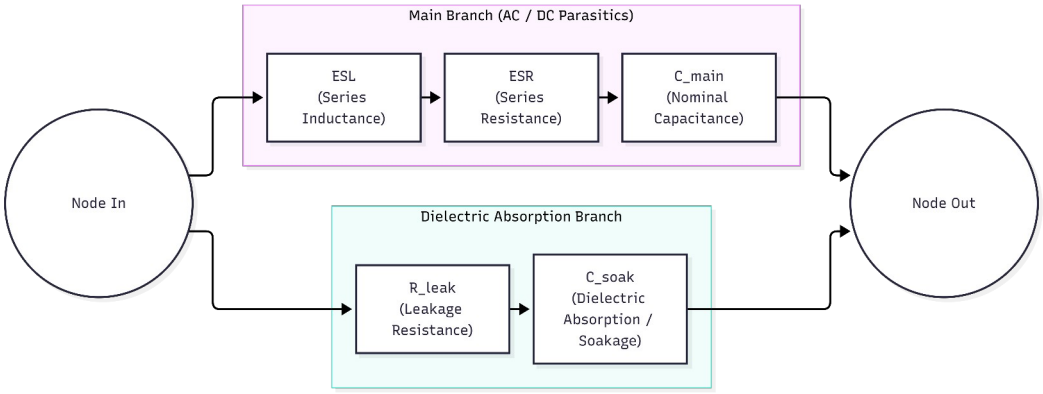
High-level design of an analog circuit requires modeling real-world imperfections before physical assembly.

4.1. Capacitors and Dielectric Soakage

In the integrator nodes and feedback networks of the compensator, the use of high-dielectric-constant ceramic capacitors (X7R/Y5V type) is strictly prohibited due to their high DC voltage coefficient and high dielectric loss.

- *Effect of Dielectric Absorption:* The molecules in the dielectric retain a residual charge that they slowly release after a rapid discharge, causing integration drift and memory errors in the PID loop.
- *Approved Components:* Only polypropylene (PP) or polystyrene film capacitors for the op-amp's R-C networks, and C0G/NP0 ceramic capacitors for small capacitance values (< 10 nF).

4.2. Magnetic Cores and Hysteresis



The power inductance *L* exhibits three nonlinear phenomena:

- *Losses in the Core (Steinmetz Equation):* $P_{core} = k \cdot f^{\alpha} \cdot B^{\beta}$. At 100 kHz, the flux density ΔB must be limited to prevent thermal runaway in the core.
- *Skin Effect and Proximity Effect:* At 100 kHz, the depth of current penetration in copper is $\delta \approx 0.21$ mm. Multifilament yarn is required (Litz) or flat windings to minimize AC resistance (R_{AC}).

4.3. Error Operational Amplifier: Input Imperfections

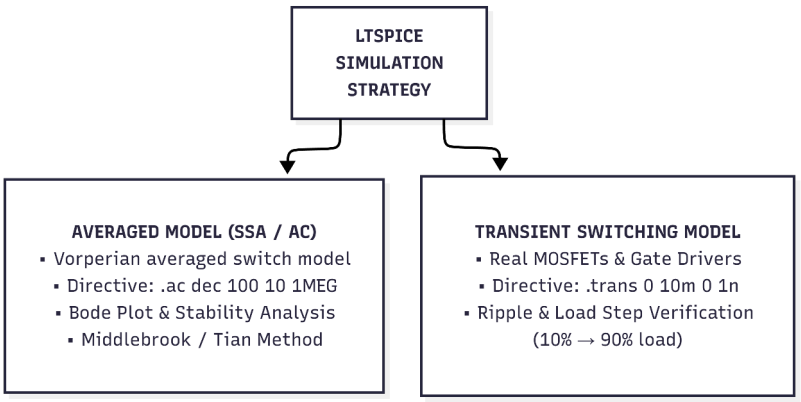
The op-amp for the analog compensator should be selected by evaluating:

- *Input Bias Current (I_b):* A current $I_b > 100$ pA when continuously injected into the high-impedance summing node, it will charge the integration capacitor C_f, causing the output to drift toward saturation of the power rails.
  - *Requirement:* $I_b > 10$ pA (JFET or ultra-low-leakage CMOS technology, e.g., OPA140 or LMC662).

- *Product-Bandwidth Product (GBW):*
  - *Requirement:* GBW ≥ 100 x f_{sw} = 10 Mhz to ensure that the open-loop gain A_{OL} the op-amp's phase response does not degrade at the desired crossover frequency (10 kHz).

5. Pre-Layout Simulation Strategy in Ltspice

The verification process in LTspice is divided into two distinct stages:



5.1. Signal Injection Protocol (Tian/Middlebrook Method)

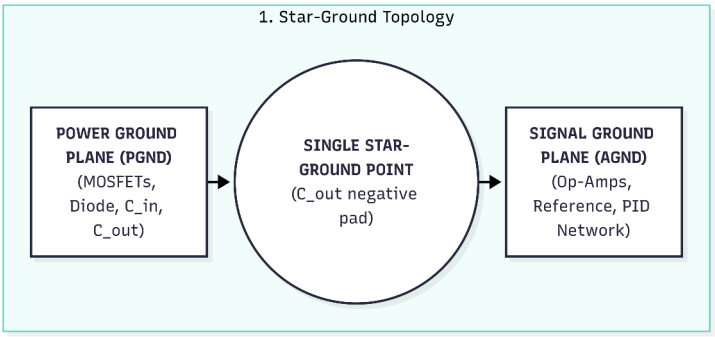
To measure closed-loop gain $T(s) = A(s) \cdot \beta(s)$ Without disrupting the DC operating point of the analog loop, a test AC voltage source is inserted (V_{test}) in series with the feedback network.

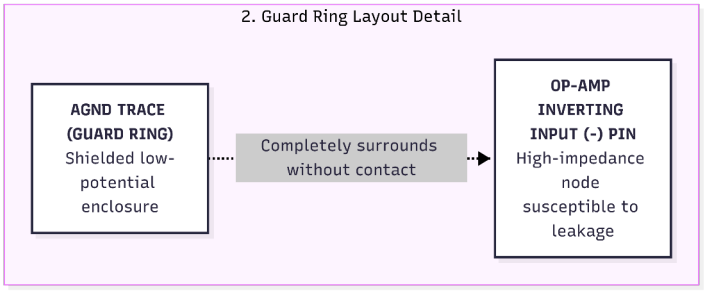
$$\text{Margen de Fase } (\Phi_m) = 180^o + \angle T(jw_{crossover}) \geq 60^o$$

$$\text{Margen de Ganancia } (A_m) = -20 \log_{10} |T(jw_{-180^o})| \geq 10 \text{ dB}$$

6. Rules for Electromagnetic Tracing

The most common failure in analog power controllers is contamination of the high-impedance summing node by radiated (dv/dt) and conducted (di/dt) noise from the power stage.





6.1. Signal Injection Protocol (Tian/Middlebrook Method)

- *Ground Separation:* The Power Ground (PGND) and Analog Ground (AGND) planes are completely separated on the PCB and are connected strictly at a single star point located beneath the negative terminal of the output capacitor C_{out}.
- *Guard Rings:* Trace the analog ground (AGND) signal around the inverting input pins (V_-) of the error op-amps to drain surface leakage currents from the PCB caused by moisture or flux residue.
- *Reducing the Switching Node (SW) Area:* The copper in the node that connects the MOSFETs to the inductor L generates intense electric fields (E-field; dv/dt ≈ 10 V/ns). Its area must be minimized and must never cross layers adjacent to the analog compensator traces.
- *Bipolar Symmetrical Power Supply:* The op-amps in the control loop will be powered by dedicated ±12 V regulated rails derived from ultra-low-noise LDO regulators (VRMS < 10 µV), independent of the 24 V input line.

7. References and Sources of Contrast

To verify and explore in greater depth the mathematical and physical concepts presented in this document, the following sources of technical reference literature are cited:

1. *Middlebrook, R. D., & Ćuk, S. (1976).* A General Unified Approach to Modeling Switching-Converter Power Stages. IEEE Power Electronics Specialists Conference (PESC).
   (A fundamental reference for the state-space averaging method).

2. *Erickson, R. W., & Maksimović, D. (2001).* Fundamentals of Power Electronics (2nd ed.). Springer Science & Business Media.
   (Chapters 7 and 8: Formal derivation of Gvd(s) and design of Type II/III compensators).

3. *Basso, C. (2012).* Designing Control Loops for Linear and Switching Power Supplies: A Tutorial Guide. Artech House.
   (A practical guide to pole-zero placement and stability margins).

4. *Analog Devices / Linear Technology.* Application Note 19: LTspice IV Circuit Notes - Middlebrook Loop Gain Measurement.
   (Signal Injection Methodology in SPICE Simulations).

5. *Texas Instruments. SLOA020A:* Precision Op-Amp Single-Supply Design Techniques
   (Analysis of bias currents Ib, offset, and guard ring design).

6. *National Aeronautics and Space Administration (NASA).* NASA SP-8018: Spacecraft Gravitational Torques—Analog Flight Computers Reference.
   (Fundamentals of Drift Mitigation in Pure Analog Integrators).